

Further Examples of R^2

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- The R^2 is the percentage of the total variation in Y “explained” by the OLS regression.
- In addition, we can show that $TSS = ESS + SSR$, where SSR denotes the sum of squared residuals. Therefore, we have

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS}, \quad (1)$$

where SSR captures the “unexplained variation”.

- $R^2 = 1$ is a perfect fit (all the data points are on the OLS regression line).
- $R^2 = 0$ means the model is explaining none of the variation in Y ; so your best guess for Y is just the same average $\bar{Y}$.

- It turns out that for the simple linear regression model with one regressor, there is a close link between R^2 and the sample correlation

$$r_{XY} = \frac{s_{XY}}{\sqrt{s_X^2 s_Y^2}}, \quad (2)$$

where s_{XY} , s_X^2 , and s_Y^2 denote the sample covariance between X and Y , the sample variance of X , and the sample variance of Y , respectively.

- Note that R^2 is a measure of the fit of the linear regression model, and r_{XY} is a measure of the linear relationship between two random variables.
- In fact, for the simple linear regression model with only one regressor, we have $R^2 = r_{XY}^2$, which is where it got its name! (just for your reference, **No Need** to recite)

- In the CA data, for example, $r_{XY} = -0.2264$.

```
. correlate str testscr
(obs=420)
```

	str	testscr
str	1.0000	
testscr	-0.2264	1.0000

- We can see that $(-0.2264)^2 = 0.0512$, which is the R^2 from the OLS regression of STR on Test Score.
- This is useful to know since it gives us some idea of what a high or low R^2 should “look like”.
- The following figure shows some typical patterns of the sample correlation.

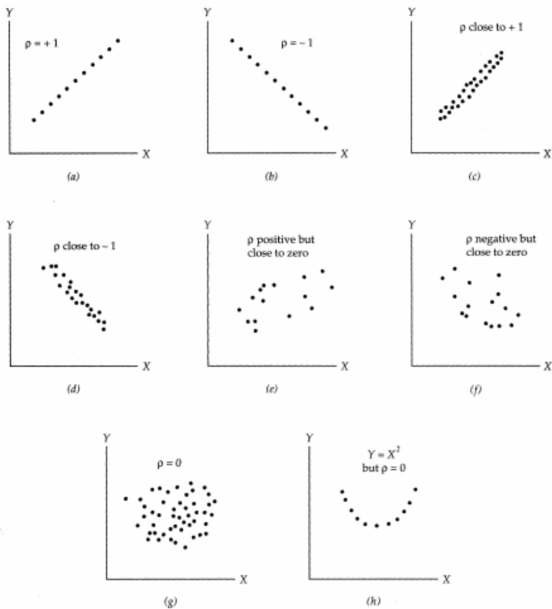
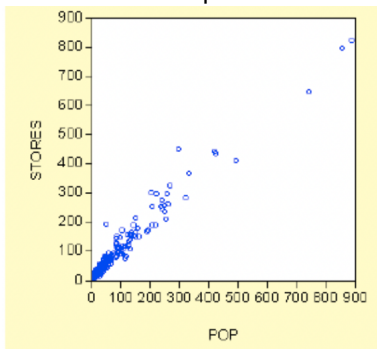


FIGURE 2-7

Some typical patterns of the correlation coefficient, ρ .

- In the lecture, we have seen what a “low” R^2 looks like. Note that this doesn’t mean that our STR regression is useless!
- Let’s look at another example. We have data on the number of supermarkets (*Stores*) (in units of supermarket stores) and population (*Pop*) for 320 Metropolitan Statistical Areas (MSAs) (in units of 10,000 people) in US.
- Here is a scatterplot of *Stores* versus *Pop*.



- It looks like a one-to-one relationship: about one store for every 10,000 people or so.
- How much of the variation in *Stores* do you think can be explained by variation in *Pop*?

```
. reg stores pop, robust
```

Regression with robust standard errors

```
Number of obs =      320
F( 1, 318) = 1362.27
Prob > F      = 0.0000
R-squared     = 0.9591
Root MSE     = 20.599
```

stores	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
pop	.9596473	.0260004	36.91	0.000	.9084927	1.010802
_cons	10.10495	1.252446	8.07	0.000	7.640823	12.56908

- The R^2 is almost 96%!